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SUPERSEDING
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MILITARY SPECIFICATION

MIXER, ROTARY TILLER, SOIL STABILIZATION, SELF-PROPELLED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a self-propelled, diesel-engine-driven, soil-stabilizing mixer.

2* APPLICABLE DOCUMENT'S

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

VV-F-800	- Fuel Oil, Diesel.
VV-G-1690	- Gasoline, Automotive, Leaded or Unleaded.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document improvement Proposal (DD Form 1426) appearing at the end Of this document or by letter.

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MIL-P-514	- Plates, Identification, Instruction and marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-T-3351	- Tractor, Full-Tracked, Low Speed; Tractor Wheeled, Industrial: and Their Attachments; Packaging of.
MIL-A-8421	- Air Transportability Requirements, General Specifications For.
MIL-E-10062	- Engines: Preparation for Shipment and Storage.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD–129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property. .
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1474	- Noise Limits for Army Materiel.

HANDBOOKS

MIL-HDBK-113	- Guide for the Selection of Lubricants, Fluids, Preservatives and Speciality Products for Use in Ground Equipment Systems.

(Copies of specifications, standards, handbooks, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents net listed in the DoDISS shall be the issue of the Non-Government documents which is current on the date of the solicitation.

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Department of Defense Materiel on Open Top Cars , section 6.

(Application for copies should be addressed to the Association of American Railroads, 50 F Street, Washington, DC 20001.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC.

Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

AMERICAN NATIONAL STANDARD INSTITUTE/AMERICAN WELDING SOCIETY (ANSI/AWS)

D 1.1 Structural Welding Code-Steel

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, N.Y, N.Y. 10018.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

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(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The mixer shall consist of a rotary soil tiller driven by a diesel engine, hydraulic traction drive, an additive pump and spray bar, a travel speed indicator, flow meter, and necessary controls all mounted on four rubber-tired wheels. The mixer shall be equipped with a rollover protective structure (ROPS) and falling object protective structure (FOPS). Dimensions and capacities shall conform to table I:

TABLE I. Dimensions

Cutting width	(Min.)	76 inches (193 cm)
Overall width	(Max. operating)	104 inches (264 cm)
Overall width	(Max. shipping)	96 inches (244 cm)
Overall length	(Max.)	25 feet 9 inches (7.85 m)
Overall height	(Max.)	102 inches (259 cm)'

3.1.1 Performance. The mixer shall be capable of performing all types of soil stabilization to include bituminous stabilization using asphalt cutbacks and emulsions, cement, lime, and choride. The mixer shall be capable of operating at depths from zero through 10-1/2 inches (266.7 mm) dependent upon soil classification and stabilization action required. The mixer cutting and mixing width shall not be less than 76 inches (193 cm). The cut shall be

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uniform depth and the bottom of the cut shall be flat. The mixer shall perform as specified herein without the use of wheel weights or hydro-inflated tires. The mixer shall operate as specified herein at altitudes up to 5,000 feet (1524 m) above sea-level. The mixer shall be capable of negotiating side slopes of 15 percent and longitudinal slopes of 27 percent without any wheel of the mixer losing contact with the ground and without the loss of fuels or lubricants. The tiller shall not be used for tilling under the slope travel requirements.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of the mixers from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish mixers conforming to this specification.

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The mixer shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the mixer may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or creatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the mixer may be newly fabricated from recovered materials to the maximum extent practicable, provided the mixer produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the mixer.

3.4 Safety. All rotating or reciprocating parts, and all parts subject to high operational temperature that are of such nature or are so located as to be, or become, a hazard to operating or maintenance personnel shall be guarded or insulated to the extent necessary to eliminate the hazard. All platforms, steps, and ladder rungs shall have antiskid surfaces. All exhausts shall be directed so as not to endanger personnel or obstruct the view of the operator.

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Fuel tanks shall be located in a mariner which will not allow spills and overflows to run onto the engine, exhaust, or electrical equipment. Electrical equipment shall be insulated, fully enclosed, or guarded. Nonfunctional sharp edges, projecting points and protruding length more than two full threads of fastening devices shall be avoided.

3.4.1 Rollover protective (ROPS) and failling object protective structure (Fops). Unless otherwise specified (see 6.2), the mixer shall be equipped with a removable, bolted down ROPS and FOPS canopy. Tine canopy shall be removable without cutting or welding. The ROPS/FOPS shall conform to SAE J104O and J231. A seat belt conforming to J386 shall be installed for use by the operator. Quick disconnect fittings (electric) shall be provided as applicable to facilitate canopy removal. A slinging eye(s) shall be provided to facilitate removal of the ROPS/FOPS.

3.5 Human factors engineering. The mixer shall be operable and maintainable by 5th through 95th percentile dressed appropriate for environments of interest (i.e. arctic, NBC, etc.) in daylight and darkness in accordance with SAE J833 and SAE J154a. Instruments shall be designed in accordance with SAE J898 and SAE J297. Controls and displays shall be provided with universal symbols in accordance with SAE J298.

3.6 Noise limits. The noise produced by the mixer shall conform with MIL-STD-1474 requirements, with the exception of MIL-STD-1474 5.2, 5.3, and 5.4, when tested in accordance with 4.5.2.14. The provision of MIL-STD-1474, 4.3 and 4.4, shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3, except the sign will state “HEARING PROTECTION REQUIRED WITHIN (specify) FEET.” The sign shall be readable at the distance indicated on the sign. Noise levels shall not exceed the requirements of the EPA nor OSHA for like equipment.

3.7 Maintenance.

3.7.1 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement with minimum removal of other major assemblies and installed attachments. Covers or plates which must be removed for component adjustment, or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be accessible and shall not drain onto any component or part. Ease of maintenance provisions shall incorporate features insuring operating clearances for handtools and lubricating equipment , and the accessibility for service operations such as adding oil to the engine crankcase and adding oil to gear cases. Access for maintainability shall be in accordance with SAE J925. Each maintenance, assembly, or disassembly operation required during testing shall be accomplished by no more than two people (except a hoist and operator may be used as necessary) using special tools furnished with the mixer and common tools. After completion of any maintenance operation, the mixer shall operate as specified herein.

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3.7.2 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Common hardware and fastening devices such as bolts, nuts, screws, and washers shall be held to a minimum number of types and sizes. All screw threads shall be in accordance with FED-STD-H28.

3.8 Chassis. All componeats of the mixer shall be mounted on a chassis of welded-steel construction. The chassis shall have four pneumatic-tired wheels and shall be provided with power-assist, automotive steering or articulated steering. Hydraulic or air-actuated brakes shall be provided on the drive wheels. The brakes shall bring the mixer to a full stop from 12 mph in not more than 25 feet. A method for attaching towing chains or cables to the front of the chassis shall be provided, In the event of power assist failure, the brakes shall be operable with a force not to exceed 70 pounds (32 Kg) and the steering shall be operable with not more than 35 pounds (16 Kg).

3.9 Engine. The manufacturer shall provide a diesel engine of the same type, size, and horsepower that is standard to the commercial product now being offered in the manufacturer's literature. The engine shall be equipped with dry type air cleaner. The fuel tank shall have a minimum capacity of 50 gallons (189.27 litre). Oil filters shall be full-flow design. The manufacturer shall supply the following gages:

(1) Engine tachometer(s).
(2) Water Temperature.
(3) Ammeter(s) or voltmeter(s).
(4) Fuel gage(s).
(5) Engine oil pressure gage(s).

The engine gages shall be visible and the engine controls shall be accessible from the operators station (platform) in accordance with provisions of 3.5. The engine shall be provided with a dipstick for measuring oil level which is marked per SAE J614b and is in an easily accessible location. The engine in the item application shall be capable of operating continuously at rated load in an ambient temperature of 120 °F (48.9 °C). Heat rejection calculations shall be furnished and will be acceptable in lieu of testing. Calculated temperatures shall not exceed the following established limits for an ambient temperature of 120 °F:

(1) Radiator top tank temperature - 210 °F. (79.38 °C)
(2) Engine oil gallery temperature - 230 °F. (86.94 °C
(3) Engine oil sump temperature - 250 °F. (94.50 °C)

3.10 Electrical system. Unless otherwise specified (see 6.2), the manufacturer shall supply a 24 volt electrical system. Wires, cables, and conductors shall be color-coded or numbered to indicate the connectors with which they mate. Long wires, cables, or bundles of cables shall be secured to the equipment chassis.

3.10.1 Lights. Unless otherwise specified (see 6.2), the mixer shall be equipped with four 24 volt flood lights (2 front, 2 rear) and positioned in such

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Lights shall be protected with a wire mesh screen having openings of not more than 1/2 inch (12.7 mm) manufactured from number 11 gage wire (2.30 mm) or larger. The operating panel shall be lighted. Floodlights and panel lights shall be incorporated into one wiring circuit.

3.10.2 Batteries. Unless otherwise specified (see 6.2), one set of batteries shall furnish the power for all electrical demands. The batteries shall be shipped dry charged. When specified a maintenance free battery or wet charge battery shall be furnished.

3.10.3 Battery cables. Battery cables between the battery and starting motors shall conform to the SAE recommended practice for heavy-duty service. The positive terminal shall be labeled with a "+" in red and the negative terminal shall be labeled with a "-" in black. The terminals shall be insulated to prevent accidental high current short.

3.10.4 Battery box. A weatherproof battery box made of fiberglass, plastic material or steel not less than 0.0747 inch (1.89738 mm) nominal thickness shall be provided for the batteries. Vents shall be provided near the top of the battery box, and drains shall be provided in the bottom. The box shall be equipped with built-in restraints constructed so that pressure will be applied to the outer battery case only. When the batteries are installed in the box with all electrical connections completed and when the battery-box lid is in the closed position, there shall be no uninsulated electrical conductor within 1 inch of any point within the battery box at which an electrical potential exist. The battery box shall be located so that the battery will be easily accessible for routine maintenance, removal, and replacement. The battery electrolyte level shall be visible when the battery filler caps are removed. The batteries shall be accessible for servicing with battery-servicing aids, such as hydrometer and a syringe. Grommets shall be provided to protect the battery lead cables. All interior surfaces of the box shall be treated with acid resistant paint. The battery box shall be provided with a pictorial wiring diagram showing proper battery connections and is visible with the battery-box lid in the open position.

3.11 Propulsion system. The propulsion system shall be capable of propelling the mixer at infinitely variable speeds from 0 through 2.3 mph (3.7 Kmph) minimum working speed with a minimum travel speed of 12 mph (19.3K mph). Direction of travel shall be controlled through a single lever. The propulsion system shall provide dynamic braking as well as motive power. The propulsion system shall be equipped with filters, coolers, and pressure relief bypasses as recommended by the component manufacturer for applications of this type. The drive system shall be independent of all other systems.

3.12 Rotor. The rotor assembly shall be mounted on dust proof, anti-friction bearings. The rotor shall be driven through a bevel gear and torque tube arrangement or by power transmission chains or hydraulically with or without a gear reduction assembly. The rotor speed and forward travel speed shall be independently variable. All gears and chains shall be enclosed. Provisions shall be made to prevent damage to any part of the rotor, or the rotor drive train, if one or more of the tines strikes an obstruction.

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3.12.1 Tines. The mixer shall be equipped with one set of tines installed on the rotor and one set of tines stored in the toolbox. The tines shall be attached to the rotor assembly by bolts, quick release positive locking devices or tines with wedge-shaped shanks that are knock in/knock out of forged type sockets. All tines furnished shall be the contractor's standard mixing and blinding tines for smooth bottom cuts, and all tines shall be interchangeable. When specified (see 6.2), four additional sets of tines with bolts, nuts, or locking devices shall be furnished and shall be packed separately.

3.12.2 Hood. The rotor assembly shall be enclosed in a hood that forms a chamber to in mixing and processing the pulverized material. The hood shall be made of steel not less than 0.1345 inch (3.4163 mm) nominal thickness. The trailing edge of the hood shall be hinged and operated hydraulically. The bottom of the hood ends shall have replaceable wearing surfaces.

3.12.3 Controls. The mixer shall be furnished with a power-driven hydraulic system to raise and lower the rotor and hood assembly and to open and close the tailboard. All controls for the hydraulic system shall be accessible from the operator's station in accordance with 3.5.

3.12.4 Spraybar. The mixer shall be equipped with a spraybar located so that liquified additives will be applied as the material is blended. The discharge angle of the spraybar shall be adjustable to permit directing the additives into the mixing chamber. The spraybar shall be equipped with nozzles that have a total capacity of not less than 175 gallons per minute (gpm) (662 litres per minute). The rate of discharge shall be adjustable for any rate between 40 gpm and 175 gpm (151 lpm and 662 lpm). The nozzles shall be attached by pipe threads. A hydraulically actuated quick-acting-type shutoff shall be provided for the spraybar, and provisions shall be made to stop the flow of material from each nozzle independently, when required.

3.13 pump and metering equipment.

3.13.1 Additive pump. The mixer shall be equipped with an additive pumping system. The pumping system shall deliver liquid asphaltic materials, asphaltic emulsions, or water to the spraybar at the rate specified in 3.12.4. The pump shall be driven by a separate engine or through a hydraulic drive. All controls for the pumping system shall be accessible from the operator's station.

3.13.2 Additive pump fittings. The pump shall be equipped with a shutoff valve, a flowmeter valve, a flowmeter indicating the rate of discharge in gallons per minute, and a volumeter that shall record the total gallons. A strainer shall be provided in the inlet to the pump. The pump inlet shall be fitted with a 16-foot length of 3-inch inside diameter rubber hose. The hose shall be wire reinforced and shall withstand pumping asphaltic materials at a temperature of 200 °F (93.3 °C). The pump inlet shall be equipped with fittings end components for a 3-inch quick-disconnect coupling. Each end of the hose shall be equipped with hose end components for 3-inch quick-disconnecting couplings. A dust cap shall be provided for the inlet to the pump and for each end of the hose. Each dust cap shall be secured by a chain. Provision shall be

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made to prevent the hose from dragging on the ground when the hose is attached between a supply truck and the mixer. Provisions shall be made to securely store the hose on the mixer when the hose is not in use.

3.13.3 Speed indicator. A speedometer indicating the traveling speed in feet per minute shall be provided. The speedometer dial shall be located so that it is visible from the operator's station. The indicator may be driven by one of the mixer wheels or by a ground contact wheel.

3.13.4 Ground contact wheel. When the speedometer is driven by a ground contact wheel, the wheel shall be rubber-tired. A control for raising and lowering the wheel shall be provided and shall be accessible from the operator's station. A means shall be provided to lock the wheel in the raised position. When the wheel is in the raised position, the clearance between the wheel and ground shall be not less than that for other components of the mixer.

3.13.5 Distribution chart. The mixer shall be provided with a chart correlating the forward working speeds of the mixer with the rates at which liquids will be discharged into the mixing chamber. The chart shall be calibrated in increments of not more than 0.25 gallon per square yard, and shall indicate the rates of application from not more than 0.50 gallon to not less than 4 gallons per square yard. The chart shall be depicted on a plate as specified in 3.2.5.

3.13.6 Pump engine (when provided see 3.13.1). Unless otherwise specified (see 6.2), the engine shall be diesel using power for electrical cranking and fuel for operation from the mixer batteries and fuel tank. When specified (see 6.2), an air cooled gasoline engine shall be provided with a fuel tank with a capacity for not less than 8 hours rated load operation and an electrical cranking system powered by the batteries for the mixer engine. The following instruments shall be provided:

(1) Lubricating oil pressure gage (if applicable).
(2) Hourmeter.
(3) Tachometer.

Instruments and controls for the pump engine shall be in accordance with human factors engineering provisions of 3.5.

3.13.6.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the pump at rated-output conditions will not require horsepower in excess of the continuous brake horsepower rating of the engine for the applicable speed as established by the engine manufacturer.

3.14 Parking brake. The mixer shall be equipped with a parking brake that shall prevent the mixer from rolling free on a 27 percent grade that has a smooth surface.

3.15 Wheels and rims. The mixer wheels shall be the contractor's standard. Rims shall conform to recommendations of the Tire and Rim Association Yearbook.

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3.16 Tires and tubes. Tires shall conform to the recommendations of the Tire and Rim Association Yearbook. Tires shall be high flotation. All tires shall nave a load rating of not less than the actual load imposed on the tire. Tires shall be of not less than 100 level quality.

3.17 Operator's platform. The mixer shall be provided with an operator's platform and steps made of steel that have a skidproof surface. Grab-rails or handholds shall be provided to assist mounting and dismounting. A padded seat with a weatherproof covering and a pack support shall be installed on the operator's platform. The operator's station shall be located on the platform where the operator can sit or stand to operate the mixer.

3.18 Toolbox. The mixer shall be provided with a weatherproof toolbox or toolboxes made of steel not less than 0.0747 inch (1.89738 mm) nominal thickness. The toolbox or toolboxes shall provide storage for the replacement set of tines, the technical publications, and the maintenance tools. The toolbox lid shall be mounted on piano binges. The lid shall open not less than 90 degrees and shall remain in the fully-open position either by its own weight or by holding devices. Each lid shall be provided with two hook-type fasteners. The fasteners shall be attached to the box by either bolts or rivets. A hasp that will accommodate a padlock that has a 3/8-inch (9.525 mm) diameter shackle shall be provided for each toolbox lid.

3.19 Lubrication. All surfaces requiring lubrication shall be provided with a means of lubricating.

3.19.1 Lubricants. Unless otherwise specified (see 6.2), the procedure for the selection of lubricants shall be in accordance with section 5 of MIL-STD-838. Lubricants selected shall be in accordance with chapter 2 of MIL-HDBK-113. When the specification of the lubricant selected includes a requirement for a qualified products list (QPL) the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.19.2 Lubrication fitting. Fittings shall be located in a protected position and shall be accessible to a grease gun. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts.

3.19.3 Pressure-release device. An automatic pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.19.4 Grease lubricaiton. Unless otherwise specified (see 6.2), grease lubrication shall be military grease. The mixer shall be assembled, run-in, tested, and delivered with this grease. The mixer shall be tagged in a conspicuous place to indicate what type of military grease has been used.

3.19.5 Oil lubrication. Unless otherwise specified for packaging, all components shall be filled to the operating level with military type oils. Each component shall be tagged in a conspicuous place to indicate the temperature range, grade and type or lubricant used.

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3.20 Electromagnetic interference. Unless otherwise specified (see 5.2), the electromagnetic interference emission and susceptibility characteristics of the mixer shall conform to SAE J551, devices category.

3.21 Lifting attachments. Unless otherwise specified (see 6.2), permanently affixed attachments that enable the mixer to be lifted in its normal travel position shall be provided. Each attachment shall meet the requirements specified in MIL-STD-209 for type 11 equipment, class 1 or 3 provisions. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.22.

3.22 Tiedown attachments. Unless otherwise specified (see 6.2), permanently affixed attachments that permit fastening mixer to the floor or deck of a transportation medium shall be provided. The tiedown attachments shall meet the requirements specified in MIL-STD-209 for type II equipment, class 2 or 3 provisions. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.21.

3.23 Shipping data plates. Shipping data plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and shall show the silhouette of the mixer in transport position indicating the center of gravity and the location and capacity of the lifting and tiedown attachments. The recommended lifting and tiedown arrangement shall be included on these plates. The data plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.24 Identification marking. The mixer shall be identified in accordance with MIL-STD-130. The markings shall be applied to the mixer on a plate conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.25 Control and display identification, safety and warning plates. A plate shall be furnished for each switch, gage, and control and at all points requiring attention for safe and efficient operation of the mixer. The plates shall be labeled by lettering or other appropriate marking with information such as function, direction each control is to move, neutral position of each control, safety instructions, and other warnings or instructions, as necessary. In addition, the plates shall be in accordance with 3.5. The plates shall be anodized aluminuim, sunken-etched or raised-relief. Plates that describe safety measures or that warn against operations decremental to the mixer shall be distinguished by the words “CAUTION” or “WARNING” in red lettering unless specified differently by applicable documents. The plates shall be attached by screws, bolts, or rivets in a conspicuously protected location,

3.26 Treatment and painting. The portions of the mixer normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage green 383 conforming to MIL-C-46168.

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When specified (see 6.2), the mixer shall then be overcoated in accordance with the Government furnished camouflage patterns with MIL-c-46168 top coat of the colors specified in the camouflage patterns except the first article may be camouflage green 383.

3.27 Stenciling. The gross weight of each mixer shall be stenciled on each side of the mixer. The prescribed tire pressure in accordance with the Tire and Rim Association Yearbook shall be stenciled "TP (specify lbs.)" on each fender. "LIFT" and "TIEDOWN", as applicable, shall be stenciled adjacent to their respective tiedown and lifting attachments. All stenciling shall be in block-type letters not less than 1 inch (25.4 mm) high, and shall be black in color.

3.28 Workmanship.

3.28.1 Steel fabrication. Steel used in the fabrication of the mixer shall be free from kinks and sharp bends. Steel plate shall not be warped. The straightening of material shall be done by methods that will not cause deformation to the metal. Burned surfaces of flame-cut material shall be ground or machined to remove slag and cooling checks. All bends of a major character shall be made with metal dies or fixtures in order to insure uniformity of size and shape.

3.28.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washer or lockwashers shall be provided and all bolts, nuts, and screws shall be tight.

3.28.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full, concentric with the rivet holes, and in full contact with the surface of the member.

3.28.4 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Weldments, including weld beads, shall be of homogeneous appearance and form without spattering or other irregularities, complying with the size, configuration and other dimensional requirements to develop the full strength of the parts joined by welds. No cracks of any type are permitted. All welding is to be in accordance with AWS D1.1. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to service loading.

3.28.5 Surfaces. All parts, components, and assemblies including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be cleaned and free of sand, dirt, fins, pits, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth , and all edges shall be rounded or beveled.

3.29 Transportability,

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3.29.1 Rail. Unless otherwise specified (see 6.2), the mixer shall be rail transportable by wood-decked, open-top railcars. The overall dimensions of the mixer with the exhaust and air cleaner stacks removed, shall meet the Passe-Partout International (PPI) requirements (see figure 1) for rail shipment. When specified {see 6.2), mixers shall successfully complete the test described in 4.5,2.18, test procedure for rail shipment.

3.29.2 Air. Unless otherwise specified (see 6.2), the mixer shall be air transportable by C-130, C-141, and C-5 aircraft and meet the requirements of MIL-A-8421. Disassembly to conform to this requirement shall be permitted provided that such disassembly and reassembly can each be accomplished by one man within 1 hour. The envelope requirements of SAE J154 are not waived for air transportability, even though the ROPS/FOPS may be removed (vehicle must be capable of being driven onto aircraft).

3.29.3 Water. Unless otherwise specified (see 6.2), the mixer shall be water tansportable by all general cargo ships (breakbulk, roll-on/roll-off, LASH, and SEABEE) and by the LARC XV and larger Army lighterage.

3.29.4 Highway. Unless otherwise specified (see 6.2), the mixer shall be highway transportable at its gross vehicle weight (GVW) without exceeding the payload capacities of the vehicle combination providing transport.

3.30 Vehicle weight classification. The contractor shall complete the Government-furnished vehicle weight classification forms as specified (see 6.5).

3.31 Camouflage patterns data. When specified (see 6.2.1), the contractor shall generate camouflage pattern data.

3.32 Government furnished property. When specified (see 6.2.1), the following property will be furnished by the Government (see 6.7):

Item No.	Description	Identification	Quantity
1	Pattern drawings	Camouflage paint patterns	As required

4. QUAILITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all produces or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling' in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First Article.

4.3.1 Examination. The mixer shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The mixer shall be tested as specified in 4.5.2, through 4.5.2.19. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each mixer shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each mixer shall be tested s specified in 4.5.2. Any damage, malfunction of the controls, instruments, brakes, lights, engine, pump, or excessive vibration or failure of any test shall be cause for rejection.

4.5 Inspection procedure

4.5.1 Examination. The mixer shall be examined for the following defects:

101. Any dimension not as specified.
102. Assembly not complete.
103. Material not as specified.
104. Safety devices missing or not as specified.
105. Human factors engineering not as specified.
106. Ease of maintenance not as specified.
107. Interchangeability not as specified.
108. Chassis not as specified.
109. Engine instrument not as specified.
110. Lights not as specified.
111. Battery box not as specified.

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112. Propulsion system not as specified.
113. Rotor not as specified.
114. Tines not as specified.
115. Hood not as specified.
116. Hydraulic system and controls not as specified.
117. Spraybar not as specified.
118. Additive pumps not as specified.
119. Speed indicator not as specified.
120. Distribution chart not as specified.
121. Pump engine and instruments not as specified.
122. Wheels and rims not as specified.
123. Tires and tubes not as specified.
124. Operator's platform not as specified.
125. Toolbox not as specified.
126. Lifting attachments not as specified.
127. Tiedown attachments not as specified.
128. Shipping data plates not as specified.
129. Identification marking not as specified.
130. Control identification, safety and warning plates not as specified.
131. Treatment and painting not as specified.
132. Stenciling not as specified.
133. Workmanship not as specified.
134. Lubricants riot as specified.
135* Paint *color* not as specified.
136. Camouflage pattern data not provided as specified
137. Materials not resistant *to* corrosion and deterioration, or or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
138. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other,
139. Contractor does not have documentation available for identification of material, material finishes, or treatment.
140. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the mixers.
141. Steel fabrication not as specified.
142. Bolted connections not as specified.
143. Riveted connections not as specified.
144. Welding not as specified.
145. Transportability not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The mixer shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test. The diesel engine shall be operated on diesel fuel conforming to

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VV-F-800 and shall be lubricated with oil conforming to MIL-L-2104. The gasoline powered pump engine, if provided, shall be operated on conforming to MIL-G-3056 or VV-G-1690, and shall be lubricated with oil conforming to MIL-L-2104.

4.5.2.2 Operational. Operate the mixer to pulverize subsoil in a level area not less than 150 feet long by not less than 8 feet wide to a depth of not less than 10-1/2 inches. After the mixer has completed one pass over the test area, determine the width and depth of the cut and flatness of the bottom of tbe cut. A cut that is less than 76 inches wide, 10-1/2 inches deep or uneven bottom of the cut shall constitute failure of this test.

4,5.2.3 Slope travel. With the tiller in the raised travel position, maneuver the mixer on side slopes of not less than 15 percent grade and longitudinal slopes of not less than 27 percent grade. Inability of the mixer to negotiate the slopes or any wheel of the mixer losing concact with the ground or any loss of fuels or lubricants shall constitute failure of this test.

4.5.2.4 Pulverizing subsoil. Operate the mixer for not less than 10 hours to pulverize compacted subsoil to the maximum digging depth. Speed shall be matched to the conditions encountered. When the test is conducted in an area where repeated passes over the same area are required, the soil shall be compacted to a density that will prevent free flow, and the uncut material will be stable. Inability of the mixer to maintain a minimum travel speed, a mimimum working depth of 8 inches (203.2 mm) material not compacted between passes, any malfunction of the mixer, except for breaking of the tines and drive line shear pins when bitting an obstruction shall constitute failure of this test. The contractor may, at his option, conduct this test on a commercial construction facility providing all other conditions specified herein are met.

4.5.2.5 Blending soil and cement. Operate the mixer blending soil, cement, and water (using the additive system on the mixer), with pulverized soil and aggregate using a suitable mix for secondary roads in an area not less than 150 feet long by not less than 12 feet wide to a depth of not less than 6 inches. Unless otherwise specified, the cement added shall not be less than 5 percent by weight of the total mix. The contractor may at his option use a commercial construction project having higher engineering standards and requirements than that specified herein, however, all requirements of this specification, and all failure criteria shall apply regardless of the project or job conditions being utilized. Inspect the blended area at 20-foot (6.096 m) intervals during the blending opeations to determine if the soil, aggregate, and cement have been uniformly blended to a depth of 6 inches (152.4 mm). Evidence that the mix is not thoroughly blended or any malfunction of the mixer or any component shall constitute failure of this test.

4.5.2.6 Blending soil and asphaltic material. Operate the mixer for 150 feet mixing asphalt material, grade MC or RC 250 (using the mixer additive systems) with pulverized soil and aggregate using a suitably designed mix for secondary road construction having a minimum thickness of 4 inches (101.6 mm). Unless otherwise specified, the asphalt material shall not be less than 5 percent by

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weight of the total mix. The contractor may, at his option, use a commercial construction project having higher engineering standards, however, all failure criteria specified herein shall apply. Evidence that the mix has not been correctly blended or any malfunction of the mixer or components shall constitute failare of this test.

4.5.2.7 Blending, aggregate. Prepare a test course by covering a 96-inch (2438 mm) wide strip 150-feet (45.72 m) long of the pulverized area with a 4-inch (101.6 mm) layer of aggregate that will pass a ~-inch sieve and be retained on a l/2-inch sieve. Set the mixer to cut 6-inches (152.4 mm) deep, and operate the mixer to blend the aggregate with pulverized material. Not more than three blending passes shall be made. Remove the material from three 6-inch (152.4 mm) deep trenches dug at 10-foot (3.948 m) intervals. Determine by visual examination, whether the aggregate has been uniformly blended with the soil. Evidence that the aggregate and soil have not been uniformly blended to a depth of 6 inches (152.4 mm) or any malfunction of the mixer shall constitute failure of this test.

4.5.2.8 Road travel. With the tiller in the raised-travel position, drive the mixer not less than 20 miles (32.187 Km) over paved roads at a speed not less than 10 mph (16.093 Kmph). Then drive the mixer for not less than 10 miles (16.093 Km) over secondary dirt roads at speeds not to exceed 10 mph (16.093 Kmph). After these operations inspect the mixer. Inability to complete these tests or any permanent deformation or damage to any component shall constitute failure of this test.

4.5.2.9 Dynamic braking.

4.5.2.9.1 Transmission braking. Use the transmission to completely stop the mixer from a forward speed of not less than 10 mph (16.093 Kmph) on a dry, level surface. Measure the distance from the point of transmission control lever application to the actual stop on three consecutive passes. Repeat above in the reverse direction. Inability of the transmission to stop the mixer within 20 feet (6.096 m) in either direction shall constitute failure of this test.

4.5.2.9.2 Drive wheel braking. Use the drive wheel brakes to completely stop the mixer from a forward speed of 12 mph (19.312K mph) and the maximum reverse speed on a dry, level surface, inability to stop within 25 feet (7.62 m) in either direction shall constitute failure of this test.

4.5.2.10 Parking brake. Place the mixer on a 15 percent grade and restrict it from rolling free by use of the parking brake only. Movement of the mixer shall constitute failure of this test.

4.5.2.11 Additive pump capacity. Operate the pump in two stages to pump liquid asphalt through spraybar. The first stage shall be at a rate of not more than 40 gpm (151.42 lpm) and the second stage at a rate of not less than 175 gpm (662.4 lpm). Liquid asphalt material, grade MC-250, heated to the spraying temperatures recommended by the manufacturer of the material shall be pumped during this test. Inability to pump the material through the spraybar at a rate of not more than 40 gpm (151.42 lpm), or at a rate of not less than 175 gpm (662.4 lpm) shall constitute failure of this test.

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4.5.2.11.1 Engine horsepower. When an internal combustion pump engine is furnished, the engine speed (rpm) and intake manifold vacuum shall be measured and recorded at 20-second intervals for duration of the pump capacity test specified in 4.5.2.11. These readings shall be compared with the engine speed (rpm) brake horsepower and intake manifold vacuum curve for the continuous brake horsepower condition as established during testing by the engine manufacturer. Inability to pump at each race specified (see 4.5.2.11) or any intake manifold vacuum reading recorded during the pump capacity test that is lower than the reading listed on the continuous horsepower curve established by the engine manufacturer at the corresponding speed (rpm) shall constitute failure of this test.

4.5.2.12 Tire load. With the fuel tanks and the radiator filled to capacity, with all the gearcases and hydraulic systems filled to capacity, and with the rotor and rotor hood raised to the normal travel height, determine the load imposed on each tire. Loads in excess of that for which the tire is rated by the Tire and Rim Association Yearbook shall constitute failure of this test.

4.5.2.13 Electromagnetic interference. Unless otherwise specified, the mixer shall be tested in accordance with SAE J551. A report tabulating the allowable maximum interference and the measured interference shall be furnished with the first article model test report. Failure to meet the requirements of SAE J551, shall constitute failure of this test.

4.5.2.14 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load. MTL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments, one increment shall include data from the noisiest position. Additionally, the noise levels at the typical operating position shall be provided as dB(A) level. Failure to meet the requirements of 3.6 shall constitute failure of this test.

4.5.2.15 Dimensional inspection. Make a dimensional check of the mixer and compare to table I and 3.29. Any dimension exceeding the dimensions in table I or nonconformance to 3.29 shall constitute failure of this test.

4.5.2.16 Test procedure of lifting provisions. Test procedures of lifting provisions shall be as follows:

a. Lift the mixer and bold in suspension in its normal travel position using slings that converge not more than 24 feet (7.3152 m) above the lowest extremity of the mixer. Determine the true angle of each sling leg relative to the vertical. Measure the clearance between each sling and the mixer. Measure eye openings and clearence dimensions of the provision. Measure the load carried by each provision.
b. Restrain the mixer by anchoring the main frame and subject each provision to a force equal to the load factor specified in MIL-STD-209 times the load it carried in a. above. An alternate

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method of test is to lift the mixer as in a. above and add weights to the main frame until the required force is obtained. Hold the mixer under either of the above loading conditions for at least 90 seconds. Weld failure or permanent deformation caused by lifting or failure to meet the requirements of MIL-STD-209, shall constitute failure of this test.

4.5.2.17 Test procedure of tiedown provisions. With the mixer anchored by means other than the provisions being tested, apply the proportion of the static loads as specified in MIL-STD-209 to each attachment. Measure dimensions of the tiedown. Failure to meet the requirements of MIL-STD-209 shall constitute failure of this test.

4.5.2.18 Test procedure for rail shipment.

Step 1. From one to five rail cars shall be positioned on a level section of track. The following conditions shall be satisfied:

The total weight of the car(s) shall be at least 114,000 kilograms (250,000 pounds).

b. The couplers between cars shall be compressed to take up any slack.

c. All of the air and hand brakes on the car(s) shall be set.

d. Any load in or on the car(s) shall be secured to prevent sliding or shifting; any movement greater than 5 cm shall be justification for retest.

e. The end of the buffer car to be struck must have a standard draft gear.

Step 2. Mount the test item on the test car. The test car shall be equipped with standard draft gears and conventional underframes. Mounting of the test item shall incorporate the standard loading and bracing method as shown in Section 6 of the Association of American Railroads (AAR) "Rules Governing the Loading of Department of Defense Materiel on Open Top Cars." No exotic or unusual tiedown methods shall be used; any non-standard loading and bracing must be approved by the Military Traffic Management Command Transportation Engineering Agency (MTMCTEA, ATTN: MTT-TR, P.O. Box 6267, Newport News, VA 23606-0276 prior to testing.

Step 3. Situate the test car between the buffer cars and the locomotive, and pull the test car at least 65 meters (200 feet) from the buffer car(s) along a level section of track (a minimum distance to achieve the required locomotive speeds).

Step 4. Position the knuckles of the buffer and test cars for coupling.

Step 5. Install a timing device to measure the test car speed (±.1 km/hr) just prior to impact with the buffer car(s). Suggested methods include electronic timing (microswitches) and radar. The use of torpedoes and a stop-watch is permissible but not recommended because of the inaccuracies involved.

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Step 6. Push the car toward the buffer car(s) and, by using the locomotive's speedometer or other means, release the test car when the desired test speed is reached, thus allowing the test car to freely impact the buffer car(s).

Step 7. Repeat step 6 until the test car(s) has impacted the buffer car three times at the same end, once each at speeds of 6.4, 9.7, and 13.0 KM/hr (4, 6, and 8 mph) +0.8, -0.0 km/hr. Reverse the test car and repeat the 13.0 km/hr impact, for a total of 4 impacts.

NOTE: Adjustment of the lading or securing mechanisms, or reconditioning of the bracing or items of securement is not allowed without a complete retest.

4.5.2.18.1 Failure criteria. Failure of the tiedowns or inability of the mixer to operate shall constitute failure of this test.

4.6 Inspection comparison. When first article inspection is specified, the Government may select mixers at any time during the contract production period and subject these mixers to the examination specified in 4.5.1 and the test specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article mixer. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison mixer to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison mixer(s) and shall be cause for refusal by the Government to continue acceptance of production mixers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at not cost to the Government on mixers previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all mixers accepted prior to the completion of inspection comparison are similary deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging. The preservation and packing, shall be examined to determine compliance with MIL-T-3351, except that a first article pack inspection shall not be required. Additionally, the packaging shall be examined for compliance with the additional requirements specified in section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

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5.1.1 Level A. The components comprising a complete mixer shall be preserved in accordance with the type 11, unboxed (mobile), level A requirements of MIL-T-3351 with the following additional requirements and exception.

5.1.1.1 Additive pump. Interior surfaces of the pump shall be coated with lubricating oil conforming to MIL-L-21260, type I, grade 10 by fogging or spraying throught the inlet opening while the pump is actuated. Excess preservative shall be drained. The shutoff valve shall be closed and the dust caps installed.

5.1.1.2 Hose. The hose shall be placed and secured on the mixer as provided for.

5.1.1.3 Spraybar. Exterior unpainted surfaces of the spraybar shall be coated with type P-1 preservative. The shuoff valves shall remain open.

5.1.1.4 Miscellaneous preservation. Preservation of components not specifically mentioned in MIL-T-3351 or herein, shall be preserved as specified for components of similar design and constriction in accordance with MIL-T-3351.

5.1.1.5 Liquid cooling system. Liquid cooling systems shall be preserved in accordance with the level A requirements of MIL-E-10062 using the water and antifreeze procedure, except the antifreeze shall conform to MIL-A-46153.

5.1.2 Level B. The components comprising a complete mixer shall be preserved in accordance with the type II, unboxed (mobile), level B requirements of MIL-T–3351 with the additional requirements and exception specified in 5.1.1.

5.1.3 Level C. Each complete mixer and all items issued with the mixer shall be suitably preserved and packed for retail distribution directly to a using customer and in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with carrier rules and regulations applicable to the mode of transportation. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function should be provided protection by the use of preservative coating, volatile corrosion inhibitors, or desiccated packs. Items requiring protection from physical damage to surface, or which are fragile by nature shall be protected by wrapping, pack compartmentization, or carbonizing of the individual item. The cooling system shall be filled with a fresh clean solution of 50 percent water and 50 percent antifreeze, conforming to MIL-A-46153, and a tag attached indicating the temperature to which the cooling system can be subjected to before freezing. The crankcase shall contain the required amount of lubricating oil. A guide shall be furnished indicating the preservation and packing materials to be removed, in detail, and any other servicing required prior to placing the mixer in operation.

5.2 Packing. Packing shall be level A, or C as specified (see 6.2).

5.2.1 Level A. Each complete mixer, preserved as specified in 5.1, shall be packed in accordance with type II, unboxed (mobile), level A requirements of MIL-T-3351.

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5.2.2 Level C. Each complete mixer, preserved as specified in 5.1, shall be prepared for unboxed (mobile) shipment in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special or other identification marking required by the contract or purchase order (see 6.2), each container shall be marked in accordance with MIL-STD-129.

5.4 First article. A first article pack shall not be required.

6. NOTES

6.1 Intended use. The mixer is intended to be used for mixing and blending stabilizing materials with subgrade soils in preparation of roads and airfields.

6.2 Ordering data.

6.2.1 Acquisition requirement. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. When first article model testing is not required (see 3.2).
c* Time frame required for submission of first article {see 3.2).
d. When the Government will conduct any or all of the preproduction model testing and examination. When the Government will conduct some, but not all of the first article examination and test, the contracting officer should specify which examinations and tests will be conducted by the contractor (see 3.2).
e. When other than a removable ROPS/FOPS canopy is required (see 3.4.1).
f. When other than 24 volts are required (see 3.10 and 3.10.1).
g. When other than a dry-charge battery is required (see 3.10.2).
h. When four additional tines shall be furnished (see 3.12.1).
i. Type of pump drive required (see 3.13.1 and 3.16.6).
j Lubricant required (see 3.19.1, 3.19.4 and 3.19.5).
k. When electromagnetic interference requirements are other than specified (see 3.20).
1. When lifting attachments are not required (see 3.21).
m. When tiedown attachments are not required (see 3.22).
n. Color required when other than specified (see 3.26).
o. When mixer shall be overcoated with Government furnished camouflage patterns (see 3.26).
p. Transportability requirements (see 3.29.1 thru 3.29.4).
q. Time frame for camouflage pattern data when required (see 3.31).
r. Degree of preservation and packing required (see 5.1 and 5.2).
s. Any special marking (see 5.3),

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD For 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410.6 are invoked and the DD Form 1423 is not used, the data specified below shall -be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No	Data requirements	Applicable DID NO.
a. 3.31	Color photograph prints	DI-MISC-80192
b. 3.31	Camouflage line art data	DI-MISC-80176

6.3 First article. When a first article inspection is required, the item should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Provisioning. The contracting officer should include provisioning requirements for the repair parts and maintenance tools as necessary (including any special tools) and instruction on the shipment of the mixers.

6.5 Vehicle weight classification. The contracting officer should arrange to furnish vehicle classification forms to the contractor and request the return of the completed forms to the Government. The information thus obtained should be utilized to determine the weight classification number(s) for the mixer. The contractor should then be requested by the contracting officer to apply the weight classification number(s) to the mixer. The use of a Government-furnished sign kit conforming to MIL-S-40626 should be utilized where applicable (see 3.30).

6.6 Lubricants. MIL-STD-838, Lubrication of Military Equipment, prescribes the policy for using specifications-type products wherever possible and provides specific requirements for potential use of non-standard proprietary products.

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MIL-STD-838 is implemented by MIL-HDBK-113, Guide for the Selection of Lubricants, Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development and Engineering Center (ATTN : STRBE-VF), Fort Belvoir, VA 22060-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.7 Government-furnished property. The contracting officer sbould arrange to furnish the property listed in 3.32.

6.8 Subject term (Key word) listing.

Hydrostatic stabilizer
Mixer
Pulvi-mixer
Rotary tiller
Soil stabilizer
Soil tiller

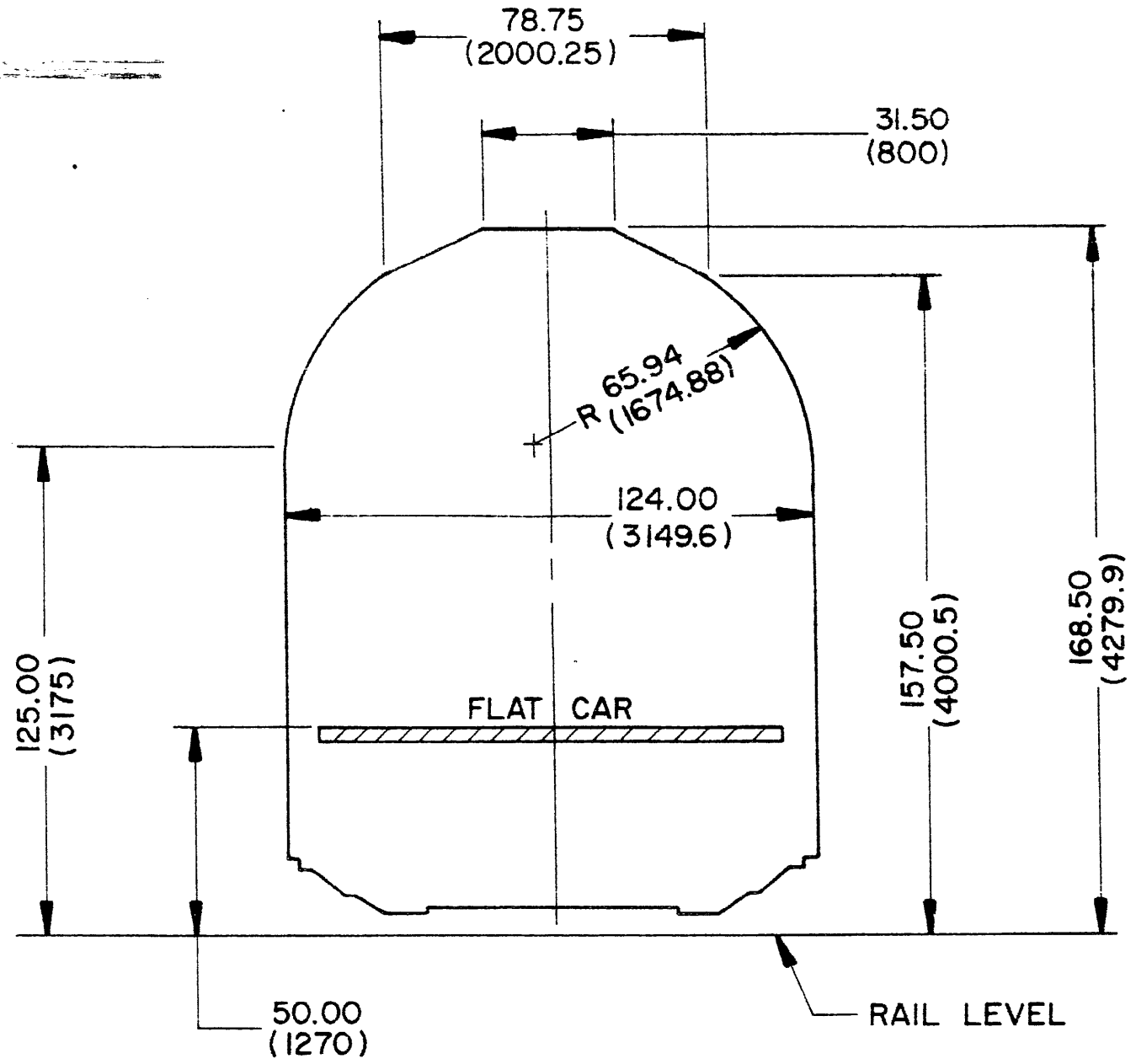
6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME
Navy - YD

Preparing activity:
Army - ME

Project 3895-0283

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NOTE: DIMENSIONS SHOWN: INCHES (MILLIMETERS)

FIGURE 2. Passe-partout international (PPI) gauge.

X-4093B

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-13081F	Mixer, Rotary Tiller, Soil Stabilication, Self-Propelled

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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